Dense-Data Local Quintic Radial-Basis-Function Surrogates for Peng–Robinson Real-Gas Thermodynamics (`realgas-ml` v0.3.0)

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Department of Chemical Engineering · Chemical Engineering Thermodynamics Project · November 2026 · GitHub Repository: `realgas_ml`

1 Executive Summary and Thermodynamic Formulation

The `realgas-ml` (v0.3.0) software package (`pyproject.toml`, `README.md`) implements a self-contained thermodynamic calculation engine and local kernel regression framework that predicts the pressure–volume–temperature (PVT), caloric, and phase-equilibrium departure properties of 15 pure supercritical fluids. Rather than relying on external runtime thermophysical libraries such as CoolProp or Cantera, the project synthesizes 65,536 analytical states per fluid (983,040 states in total) directly from the classical Peng and Robinson (1976) two-constant cubic equation of state (PR76) coupled with continuous NASA 7-coefficient ideal-gas heat-capacity polynomials. Each fluid is represented by a four-output local quintic radial-basis-function (`LocalRBFRegressor`) surrogate trained over a normalized temperature and logarithmic-pressure grid.

In the analytical solver `realgas_ml/peng_robinson.py`, pressure P (Pa) is expressed as a function of absolute temperature T (K) and molar volume v ($\mathrm{m^3\,mol^{-1}}$) via the Peng–Robinson (1976) equation of state:

$$P(T,v) = \frac{RT}{v-b} - \frac{a(T)}{v^2+2bv-b^2}, \qquad a(T) = a_c\left[1+\kappa\left(1-\sqrt{\frac{T}{T_c}}\right)\right]^2, \qquad b = 0.07780\,\frac{RT_c}{P_c} \tag{1}$$

where $R = 8.31446261815324\ \mathrm{J\,mol^{-1}\,K^{-1}}$ is the molar gas constant, $a_c = 0.45724\,R^2T_c^2/P_c$ denotes the critical attraction parameter, T_c and P_c represent the critical temperature and critical pressure, and $\kappa = 0.37464 + 1.54226\,\omega - 0.26992\,\omega^2$ is the original PR76 acentric polynomial in terms of Pitzer's acentric factor ω. Differentiating $a(T)$ analytically with respect to temperature yields the first derivative required for caloric departure evaluations, $\frac{da}{dT} = -\frac{a_c\,\kappa\,g(T)}{T_c\sqrt{T/T_c}}$, where $g(T) = 1+\kappa(1-\sqrt{T/T_c})$.

Defining the dimensionless attraction and covolume groups $A = \frac{a(T)P}{R^2T^2}$ and $B = \frac{bP}{RT}$, Equation (1) is recast into the canonical cubic polynomial for the compressibility factor $Z = \frac{Pv}{RT}$:

$$Z^3 + (B-1)Z^2 + (A-3B^2-2B)Z + (-AB+B^2+B^3) = 0 \tag{2}$$

The function `cubic_z(A, B)` in `peng_robinson.py` solves Equation (2) at each state using companion-matrix eigenvalues (`numpy.roots`), filters physically admissible roots satisfying $|\mathrm{Im}(r)| < 10^{-9}$ and $\mathrm{Re}(r) > B$ (enforcing $v > b$), and selects the largest real root to isolate the single-phase gas branch.

To evaluate real-gas caloric and chemical-potential departures (Maginn, CBE 20260), the solver defines the logarithmic factor L:

$$L = \ln\left[\frac{Z+(1+\sqrt{2})B}{Z+(1-\sqrt{2})B}\right] = \mathrm{log1p}\left[\frac{2\sqrt{2}\,B}{Z+(1-\sqrt{2})B}\right] \tag{3}$$

Evaluating L via `np.log1p` in `peng_robinson.py` eliminates floating-point subtraction of nearly equal logarithms in the low-density limit ($P \to 1$ bar, $B \ll 1$, $Z \to 1$), preserving machine precision across all 15 fluids. The molar enthalpy departure h_{dep}, molar entropy departure s_{dep}, logarithmic fugacity coefficient $\ln\phi$, and fugacity f are computed analytically as:

$$h_{\mathrm{dep}}(T,P) = RT(Z-1) + \frac{T\frac{da}{dT} - a(T)}{2\sqrt{2}\,b}\,L, \qquad s_{\mathrm{dep}}(T,P) = R\ln(Z-B) + \frac{\frac{da}{dT}}{2\sqrt{2}\,b}\,L \tag{4}$$

$$\ln\phi(T,P) = Z - 1 - \ln(Z-B) - \frac{A}{2\sqrt{2}\,B}\,L, \qquad f(T,P) = \phi(T,P)\,P \tag{5}$$

Ideal-gas caloric contributions $h_{\mathrm{ig}}(T)$ and $s_{\mathrm{ig}}(T)$ are obtained in `ideal_integrals(T, info)` by integrating the five heat-capacity coefficients $(a_0,\ldots,a_4)$ of the NASA 7-coefficient polynomial $C_p^\circ(T)/R = \sum_{k=0}^{4} a_k T^k$ from the package reference state $T_{\mathrm{ref}} = 300.0$ K and $P_{\mathrm{ref}} = 10^5$ Pa (1 bar), where $h_{\mathrm{ig}}(T_{\mathrm{ref}}) = 0$ and $s_{\mathrm{ig}}(T_{\mathrm{ref}}, P_{\mathrm{ref}}) = 0$:

$$\tilde{h}(T) = R\left(a_0T + \frac{a_1}{2}T^2 + \frac{a_2}{3}T^3 + \frac{a_3}{4}T^4 + \frac{a_4}{5}T^5\right), \quad \tilde{s}(T) = R\left(a_0\ln T + a_1T + \frac{a_2}{2}T^2 + \frac{a_3}{3}T^3 + \frac{a_4}{4}T^4\right) \tag{6}$$

The standard formation and third-law integration constants (a_5, a_6) are intentionally omitted, and piecewise polynomial sub-intervals ($[200, 1000]$ K and $[1000, 6000]$ K, or $[300, 1000, 5000]$ K for SO_2) are accumulated via clamped bounds (`np.clip`) to enforce exact C^0 continuity across the 1000 K midpoint junction (essential for H_2O, whose domain extends to 1100.06 K). Adding the real-gas departures yields total molar enthalpy $h(T,P) = h_{\mathrm{ig}}(T) + h_{\mathrm{dep}}(T,P)$ and total molar entropy $s(T,P) = s_{\mathrm{ig}}(T) - R\ln(P/P_{\mathrm{ref}}) + s_{\mathrm{dep}}(T,P)$. Because $h_{\mathrm{dep}}(300\ \mathrm{K}, 1\ \mathrm{bar}) \neq 0$, real-gas h and s are non-zero at the ideal-gas reference state.

2 Fluid Domain Characterization and Thermophysical Provenance

The dataset specification in `realgas_ml/data/metadata.json` covers 15 pure fluids spanning monatomic noble gases (He, Ne, Ar), homonuclear and heteronuclear diatomics (H_2, N_2, O_2, CO), linear and bent triatomics (CO_2, SO_2, H_2O), polar associated hydrides (NH_3), and the C_1–C_4 homologous n-alkane series (CH_4, C_2H_6, C_3H_8, n-C_4H_{10}). Static NASA7 polynomial arrays are sourced from Cantera 3.2.0 `nasa_gas.yaml` (`SHA-256: 4de6199d...8667e4db`, derived from NASA TM-4513) under the BSD 3-Clause license preserved in `THIRD_PARTY_NOTICES.txt`. Critical constants (T_c, P_c, ω, M) are transcribed from Cantera 3.2.0 `critical-properties.yaml` (referencing Lemmon et al., 2022, NIST WebBook SRD 69; Yaws, 2001; and Poling et al.) for nine species, and from static CoolProp fluid documentation tables for Helium, Argon, Neon, Ammonia, SulfurDioxide, and Water.

Table 1 reveals the physical design of the state space. Every fluid shares the pressure domain $P \in [10^5, 5\times10^6]$ Pa (1–50 bar), while the minimum temperature satisfies $T_{\min} = \max(300.0\ \mathrm{K}, 1.10\,T_c)$. Enforcing $T \geq 1.10\,T_c$ guarantees that every state lies

Table 1: Thermophysical critical constants, operating domains ($P \in [1.0, 50.0]$ bar for all fluids), reduced parameter bounds, and data provenance for all 15 pure fluids in `metadata.json`.

Fluid Name	Species	M (g mol^{-1})	T_c (K)	P_c (bar)	ω (—)	$T_{\min}$ (K)	$T_{\max}$ (K)	$T_r = T/T_c$ Range	Critical / NASA Provenance
Helium	He	4.0026	5.1953	2.2832	−0.38354	300.00	600.00	57.74–115.49	CoolProp He / L10/90
Hydrogen	H_2	2.0159	32.9800	12.9300	−0.21700	300.00	600.00	9.10–18.19	Cantera (Poling) / TPIS78
Neon	Ne	20.1790	44.4000	26.6163	−0.03549	300.00	600.00	6.76–13.51	CoolProp Ne / L10/90
Nitrogen	N_2	28.0135	126.2000	33.9000	0.03800	300.00	600.00	2.38–4.75	Cantera (NIST) / TPIS78
CarbonMonoxide	CO	28.0101	133.4000	35.0000	0.06600	300.00	499.00	2.25–3.74	Cantera (NIST) / TPIS79
Argon	Ar	39.9480	150.6870	48.6300	−0.00219	300.00	600.00	1.99–3.98	CoolProp Ar / L 6/88
Oxygen	O_2	31.9988	154.5800	50.4300	0.02200	300.00	600.00	1.94–3.88	Cantera (NIST) / TPIS89
Methane	CH_4	16.0428	190.7000	46.3000	0.01100	300.00	600.00	1.57–3.15	Cantera (NIST) / L 8/88
CarbonDioxide	CO_2	44.0098	304.2000	73.9000	0.22800	334.62	600.00	1.10–1.97	Cantera (Yaws) / L 7/88
Ethane	C_2H_6	30.0690	305.4000	48.8000	0.09900	335.94	600.00	1.10–1.96	Cantera (NIST) / L 8/88
Propane	C_3H_8	44.0956	369.7000	42.5000	0.15300	406.88	628.81	1.10–1.70	Cantera (NIST) / L 6/90
Ammonia	NH_3	17.0305	405.5600	113.6339	0.25569	446.12	689.45	1.10–1.70	CoolProp NH_3 / TPIS89
n-Butane	n-C_4H_{10}	58.1222	425.6000	37.6000	0.19900	468.16	574.00	1.10–1.35	Cantera (NIST) / L 6/90
SulfurDioxide	SO_2	64.0638	430.6400	78.8658	0.25613	473.70	524.00	1.10–1.22	CoolProp SO_2 / J 6/61
Water	H_2O	18.0153	647.0960	220.6400	0.34429	711.81	1100.06	1.10–1.70	CoolProp H_2O / L 8/89

strictly above the critical locus, preventing vapor–liquid phase splitting, Maxwell equal-area loops, and isothermal compressibility divergences $(\partial v/\partial P)_T \to -\infty$ at the critical point. Nevertheless, for high-T_c hydrocarbons such as *n*-Butane ($P_c = 37.60$ bar), Propane ($P_c = 42.50$ bar), and Ethane ($P_c = 48.80$ bar), the upper pressure bound of 50 bar reaches reduced pressures $P_r = P_{\max}/P_c$ of 1.33, 1.18, and 1.02, respectively, at reduced temperatures as low as $T_r = 1.10$. In this near-critical supercritical region, attractive intermolecular forces produce pronounced negative deviations from ideality ($Z \approx 0.65$–0.85) and steep gradients in h_{dep} and $\ln\phi$. Conversely, quantum and noble gases (He, H_2, Ne) operate at $T_r \in [6.76, 115.49]$, where repulsive covolume interactions dominate ($Z > 1$) and thermodynamic surfaces are nearly planar.

3 Surrogate Architecture, Coordinate Scaling, and Software Design

The surrogate pipeline implemented across `train.py`, `realgas_ml/regressor.py`, and `realgas_ml/model.py` addresses two numerical challenges inherent to real-gas property regression: multi-decade scale disparities across pressure and strict positivity constraints on compressibility and fugacity. Rather than fitting raw (T, P) coordinates, `train.py` and `model.py` map any physical state $(T, P) \in [T_{\min}, T_{\max}] \times [P_{\min}, P_{\max}]$ onto the unit square $\mathbf{x} = (x_1, x_2) \in [0, 1]^2$ via linear temperature normalization and logarithmic pressure scaling:

$$x_1(T) = \frac{T - T_{\min}}{T_{\max} - T_{\min}} \in [0, 1], \qquad x_2(P) = \frac{\ln(P/P_{\min})}{\ln(P_{\max}/P_{\min})} \in [0, 1] \tag{7}$$

Logarithmic pressure scaling serves a dual mathematical purpose. First, because the ideal-gas entropy contains the exact term $-R\ln(P/P_{\text{ref}})$, mapping $\ln P \mapsto x_2$ transforms this logarithmic singularity into an affine linear function of x_2 that is reproduced with zero error by the polynomial trend of the kernel interpolator. Second, uniform sampling in x_2 concentrates grid nodes geometrically toward lower pressures (1–10 bar) while maintaining balanced resolution across the 50-fold pressure ratio. The target vector is parameterized as $\mathbf{y} = [\ln Z,\ h,\ s,\ \ln\phi]^\top \in \mathbb{R}^4$. During inference in `GasModel.predict`, the predicted log-targets $\hat{y}_1$ and $\hat{y}_4$ are exponentiated ($Z = \exp(\hat{y}_1)$ and $\phi = \exp(\hat{y}_4)$), guaranteeing $Z > 0$ and $\phi > 0$ everywhere and reconstructing derived intensive and extensive properties for n moles via exact algebraic definitions:

$$v = \frac{ZRT}{P}, \quad V = nv, \quad \rho = \frac{M}{v}, \quad H = n\hat{y}_2, \quad S = n\hat{y}_3, \quad f = \phi P \tag{8}$$

Crucially, as verified in `tests/test_properties.py` and `tests/test_app.py`, fugacity f is an intensive state variable and remains invariant with respect to molar amount n, whereas extensive volume V, enthalpy H, and entropy S scale linearly with n.

In `realgas_ml/regressor.py`, `LocalRBFRegressor` wraps SciPy's `RBFInterpolator` within a scikit-learn `BaseEstimator` and `RegressorMixin` interface. The four target columns are first standardized to zero mean and unit variance using `StandardScaler`. For any query state $\mathbf{x} \in [0, 1]^2$, the interpolator identifies the $k = 48$ nearest training nodes $\{\mathbf{x}_{i_1}, \ldots, \mathbf{x}_{i_k}\}$ in Euclidean norm and forms a local quintic radial-basis-function expansion augmented by a quadratic polynomial of degree $m = 2$ with zero smoothing ($\lambda = 0$, exact interpolation):

$$\tilde{\mathbf{y}}(\mathbf{x}) = \sum_{j=1}^{48} \mathbf{w}_j\, \varphi\big(\|\mathbf{x} - \mathbf{x}_{i_j}\|_2\big) + \sum_{\alpha_1+\alpha_2 \le 2} \mathbf{c}_{\alpha_1,\alpha_2}\, x_1^{\alpha_1} x_2^{\alpha_2}, \qquad \varphi(r) = -r^5 \tag{9}$$

where the weights $\mathbf{w}_j \in \mathbb{R}^4$ and polynomial coefficients $\mathbf{c}_{\alpha_1,\alpha_2} \in \mathbb{R}^4$ satisfy the 48 interpolation constraints $\tilde{\mathbf{y}}(\mathbf{x}_{i_j}) = \mathbf{y}_{\text{std}}(\mathbf{x}_{i_j})$ subject to 6 discrete orthogonality conditions $\sum_{j=1}^{48} \mathbf{w}_j\, x_{i_j,1}^{\alpha_1} x_{i_j,2}^{\alpha_2} = \mathbf{0}$ for $\alpha_1 + \alpha_2 \le 2$. Whereas global RBF interpolation on $N = 65{,}536$ nodes would require factoring a dense $65{,}536 \times 65{,}536$ matrix (~ 34 GB in double precision and $\mathcal{O}(N^3) \approx 2.8 \times 10^{14}$ operations), the $k = 48$ local neighborhood reduces evaluation to a k-d tree query followed by solving a 54×54 linear system per query point. Each serialized model (`realgas_ml/data/<Gas>.joblib`, compressed at level 3) occupies only 2.63–2.66 MB on disk.

For inverse TV state specifications—where temperature T and molar volume v are known and pressure P must be recovered—`GasModel.pressure(T, v)` in `model.py` defines the residual function $g(P) = \hat{v}(T, P) - v$, verifies opposite signs at the pressure domain brackets $g(P_{\min}) \cdot g(P_{\max}) \le 0$, and executes Brent's root-finding algorithm (`scipy.optimize.brentq`) over $[10^5, 5 \times 10^6]$ Pa. Because $(\partial v/\partial P)_T < 0$ for all stable single-phase gas states, $g(P)$ is strictly monotonic and Brent's method converges to machine precision (relative error $< 10^{-8}$ in `test_properties.py`). The package exposes four user interfaces: the Python API (`example.py`), the command-line entry point `realgas-ml` (`realgas_ml/cli.py`), a bulk NPZ-to-CSV dataset exporter (`export_data.py`), and an interactive Streamlit web laboratory (`app.py`, styled via `.streamlit/config.toml` and launched on Windows via `run_app.bat`) supporting both (T, P) and inverse (T, v) modes, unit conversions (K/°C, bar/kPa/MPa/Pa), 80-point geometric-pressure trend curves, and side-by-side PR residual tables.

4 Empirical Accuracy, Convergence Scaling, and Thermodynamic Consistency

The training protocol in `train.py` employs a deterministic random generator (`np.random.default_rng(20261016 + index)`) to partition the evaluation sets independently for each fluid: (i) a full $256 \times 256 = 65{,}536$ Cartesian training grid on $[0,1]^2$; (ii) 512 uniformly distributed interior validation points used exclusively to record multi-grid learning curves at $N \in \{32^2, 64^2, 128^2, 256^2\}$; (iii) 2,048 uniformly distributed held-out interior test points; and (iv) 256 held-out domain-boundary test points (64 random points along each of the four edges $x_1 \in \{0,1\}$ and $x_2 \in \{0,1\}$). Relative accuracy for positive-definite properties (v and f) is quantified via Mean Absolute Percentage Error (MAPE) and Maximum Absolute Percentage Error (Max APE), derived directly from logarithmic residuals as $100\,|\exp(\hat{y} - y) - 1|$, while caloric accuracy for h and s (which cross zero near 300 K and 1 bar) is measured via Mean Absolute Error (MAE) and Maximum Absolute Error (Max AE).

Table 2: Held-out interior test ($N = 2{,}048$) and domain-boundary test ($N = 256$) surrogate errors against direct Peng–Robinson calculations for all 15 gases trained on the 256×256 (65,536-state) grid (extracted from `realgas_ml/data/*_metrics.json`).

	Held-Out Interior Test Split ($N = 2{,}048$ Random States)						Held-Out Domain-Boundary Split ($N = 256$ Edge States)					
Fluid (Metrics File)	v **MAPE (%)**	v **Max APE (%)**	f **MAPE (%)**	f **Max APE (%)**	h **MAE** (J/mol)	s **MAE** (J/mol K)	v **MAPE (%)**	v **Max APE (%)**	f **MAPE (%)**	f **Max APE (%)**	h **Max AE** (J/mol)	s **Max AE** (J/mol K)
Helium	3.46e-10	4.94e-08	2.80e-10	3.90e-08	1.02e-08	2.55e-10	1.23e-09	1.38e-07	9.38e-10	9.68e-08	2.19e-06	1.22e-08
Neon	5.49e-10	7.84e-08	5.66e-10	8.28e-08	1.98e-08	3.14e-10	5.25e-10	2.59e-08	5.46e-10	2.56e-08	7.37e-07	1.73e-08
Hydrogen	6.47e-10	7.74e-08	6.10e-10	7.26e-08	3.23e-08	3.32e-10	8.94e-10	6.60e-08	7.94e-10	4.95e-08	2.44e-06	2.82e-09
Water	9.96e-10	8.60e-08	9.53e-10	7.32e-08	4.53e-07	4.37e-10	1.47e-09	3.34e-08	1.14e-09	2.36e-08	6.54e-06	6.53e-09
Nitrogen	1.16e-09	3.12e-07	1.06e-09	2.46e-07	4.16e-08	4.22e-10	1.84e-09	5.90e-08	1.83e-09	5.43e-08	2.09e-06	2.23e-08
Oxygen	1.30e-09	1.90e-07	1.10e-09	1.20e-07	6.17e-08	4.57e-10	2.20e-09	9.14e-08	2.09e-09	7.17e-08	1.77e-06	1.78e-08
CarbonMonoxide	1.37e-09	2.81e-07	1.11e-09	1.72e-07	4.30e-08	1.78e-10	2.07e-09	1.98e-07	1.54e-09	1.05e-07	1.95e-05	6.15e-08
Argon	1.39e-09	2.72e-07	1.19e-09	1.78e-07	6.19e-08	3.25e-10	1.87e-09	3.31e-08	1.81e-09	3.18e-08	8.65e-07	5.36e-09
Methane	1.85e-09	1.60e-07	1.45e-09	8.28e-08	1.16e-07	5.04e-10	1.06e-08	1.37e-06	7.71e-09	8.10e-07	3.61e-05	2.00e-07
Ammonia	2.27e-09	2.14e-07	2.09e-09	1.87e-07	4.53e-07	6.38e-10	5.90e-09	5.11e-07	3.24e-09	2.50e-07	4.70e-05	9.10e-08
CarbonDioxide	4.78e-09	9.96e-07	2.71e-09	4.80e-07	3.78e-07	7.16e-10	2.47e-08	2.84e-06	9.46e-09	7.60e-07	8.98e-05	3.59e-07
SulfurDioxide	1.07e-08	1.98e-06	5.74e-09	7.47e-07	1.43e-06	2.40e-09	1.09e-08	4.29e-07	5.34e-09	1.62e-07	4.74e-05	8.64e-08
Ethane	1.73e-08	7.53e-06	5.38e-09	1.06e-06	8.64e-07	2.17e-09	5.28e-08	2.25e-06	1.39e-08	4.26e-07	1.75e-04	4.74e-07
Propane	1.85e-08	5.63e-06	4.65e-09	4.77e-07	1.09e-06	2.23e-09	4.66e-08	8.53e-07	1.09e-08	1.53e-07	6.82e-05	1.62e-07
n-Butane	4.08e-08	9.48e-06	6.99e-09	3.41e-07	2.16e-06	3.89e-09	8.28e-08	**1.04e-05**	1.12e-08	**1.50e-06**	**4.10e-04**	**8.24e-07**
Global Worst Case	**4.08e-08**	**9.48e-06**	**6.99e-09**	**1.06e-06**	**2.16e-06**	**3.89e-09**	**8.28e-08**	**1.04e-05**	**1.39e-08**	**1.50e-06**	**4.10e-04**	**8.24e-07**

Table 2 and Figure 2 demonstrate that across all 15 fluids and both held-out interior and boundary test sets, the maximum relative error on molar volume is $1.0365 \times 10^{-5}\%$ (n-Butane boundary), the maximum relative error on fugacity is $1.4985 \times 10^{-6}\%$ (n-Butane boundary), the maximum absolute error on molar enthalpy is $4.0967 \times 10^{-4}\ \mathrm{J\,mol^{-1}}$ (n-Butane boundary), and the maximum absolute error on molar entropy is $8.2417 \times 10^{-7}\ \mathrm{J\,mol^{-1}\,K^{-1}}$ (n-Butane boundary). These worst-case figures lie three to four orders of magnitude below the automated regression gates enforced in `tests/test_properties.py` (Max APE $< 0.01\%$ for v and f, Max AE $< 0.5\ \mathrm{J\,mol^{-1}}$ for h, and Max AE $< 0.005\ \mathrm{J\,mol^{-1}\,K^{-1}}$ for s).

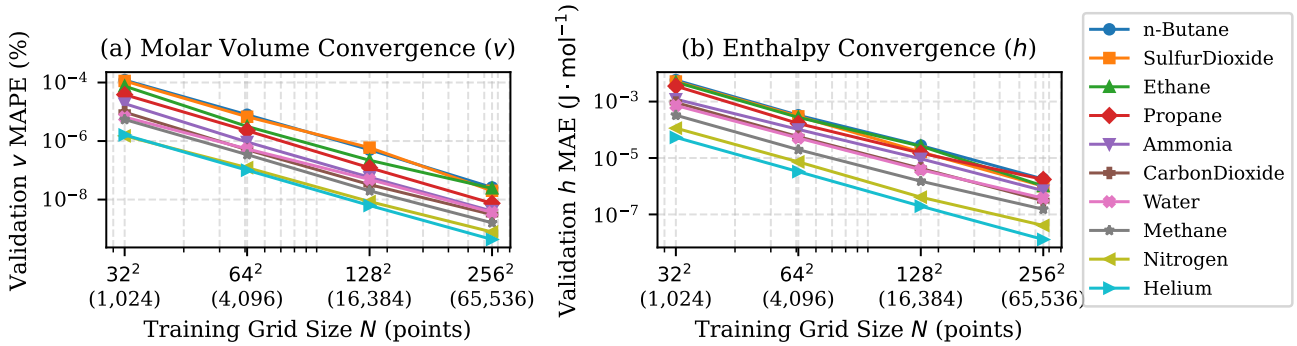


Figure 1: Validation learning curves ($N_{\mathrm{val}} = 512$) across grid resolutions $N \in \{32^2, 64^2, 128^2, 256^2\}$ for (a) molar volume MAPE and (b) molar enthalpy MAE, illustrating uniform $\sim \mathcal{O}(\Delta x^{3.8\text{–}4.1})$ algebraic error decay with mesh refinement.

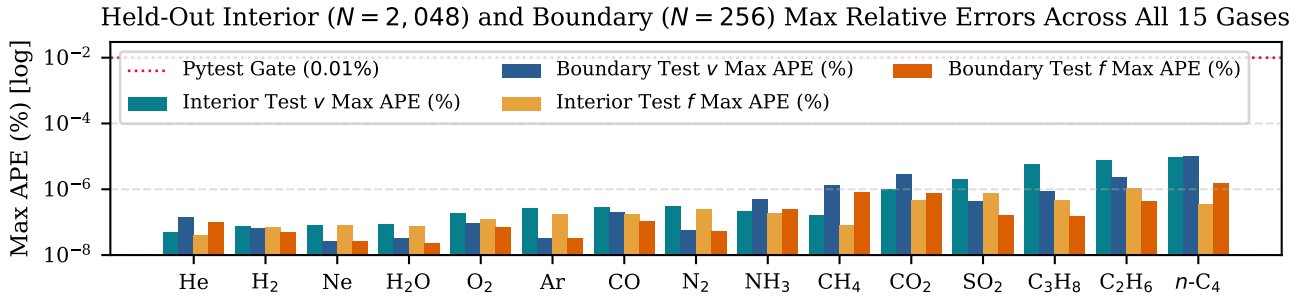


Figure 2: Comparison of held-out interior test ($N = 2{,}048$) and domain-boundary ($N = 256$) maximum relative errors on molar volume v and fugacity f across all 15 fluids at $N = 65{,}536$, benchmarked against the 0.01% acceptance gate in `test_properties.py`.

A physical hierarchy governs the error variation across fluids in Table 2. Because local quintic RBF interpolation with quadratic polynomial precision ($m = 2$) annihilates all Taylor series terms up to second order, the local truncation error at grid spacing $\Delta x = 1/(M_{\mathrm{side}} - 1)$ is proportional to high-order mixed partial derivatives $\partial^{\alpha_1+\alpha_2}\mathbf{y}/\partial x_1^{\alpha_1}\partial x_2^{\alpha_2}$ ($|\boldsymbol{\alpha}| \geq 3$). For Helium ($T_r \geq 57.7$), Neon ($T_r \geq 6.76$), and Hydrogen ($T_r \geq 9.10$), the compressibility surface $Z(T,P)$ is nearly flat and close to unity, yielding interior molar volume MAPEs between $3.46 \times 10^{-10}\%$ and $6.47 \times 10^{-10}\%$. Notably, Water exhibits similarly low interior volume MAPE ($9.96 \times 10^{-10}\%$) despite its high acentricity ($\omega = 0.3443$) and polarity because its maximum pressure of 50 bar represents a small reduced pressure $P_{\max}/P_c = 50/220.64 = 0.2266$ at $T \geq 711.81$ K, keeping supercritical steam in a weakly non-ideal regime throughout the domain. In contrast, n-Butane, Propane, and Ethane experience $P_{\max}/P_c \in [1.02, 1.33]$ near $T_{\min} = 1.10\,T_c$, where the supercritical inflection of $Z(T,P)$ across the Widom extension of the coexistence curve maximizes third- and fourth-order

derivatives. Even for these strongly non-ideal alkanes, boundary errors exceed interior errors by at most a factor of 2–5×, confirming that the 48-neighbor asymmetric stencil and degree-2 polynomial reproduction suppress Runge-phenomenon boundary oscillations.

Table 3: Validation learning-curve progression ($N_{\text{val}} = 512$) across the four training grid densities $N \in \{1{,}024,\ 4{,}096,\ 16{,}384,\ 65{,}536\}$ for all 15 gases, showing average step reduction factor $\overline{\rho}_{4\times} = \frac{1}{3}\sum_{k=1}^{3}(\text{Error}_{N_k}/\text{Error}_{N_{k+1}})$ and effective spatial convergence rate $p_{\text{eff}} = \log_2(\overline{\rho}_{4\times})$.

	Validation Molar Volume MAPE **(%) vs. Grid Size** N					**Validation Molar Enthalpy** MAE **(J mol^{-1}) vs. Grid Size** N				
Fluid	$N = 1{,}024$	$N = 4{,}096$	$N = 16{,}384$	$N = 65{,}536$	$p_{\text{eff}}(v)$	$N = 1{,}024$	$N = 4{,}096$	$N = 16{,}384$	$N = 65{,}536$	$p_{\text{eff}}(h)$
Helium	1.62e-06	1.01e-07	6.28e-09	4.33e-10	3.95	5.38e-05	3.28e-06	1.96e-07	1.29e-08	4.01
Water	1.50e-06	1.23e-07	8.55e-09	7.80e-10	3.64	7.28e-04	5.10e-05	3.83e-06	3.77e-07	3.64
Neon	1.82e-06	1.07e-07	8.70e-09	5.99e-10	3.86	7.12e-05	4.35e-06	3.56e-07	2.51e-08	3.82
Hydrogen	2.79e-06	1.83e-07	1.35e-08	1.28e-09	3.70	1.50e-04	9.14e-06	6.08e-07	4.87e-08	3.86
Oxygen	3.62e-06	2.60e-07	2.34e-08	1.88e-09	3.64	1.17e-04	6.68e-06	5.19e-07	5.63e-08	3.68
Argon	4.85e-06	2.65e-07	1.75e-08	1.73e-09	3.82	1.18e-04	9.52e-06	6.12e-07	6.33e-08	3.62
CarbonMonoxide	5.41e-06	3.83e-07	2.47e-08	2.07e-09	3.78	1.13e-04	7.06e-06	5.34e-07	5.56e-08	3.66
Nitrogen	5.45e-06	3.44e-07	2.00e-08	1.61e-09	3.91	1.14e-04	7.28e-06	4.08e-07	4.06e-08	3.82
Ammonia	6.60e-06	5.38e-07	4.87e-08	3.65e-09	3.61	1.21e-03	1.03e-04	9.37e-06	6.98e-07	3.59
Methane	9.70e-06	5.04e-07	3.24e-08	3.07e-09	3.87	3.29e-04	1.97e-05	1.51e-06	1.53e-07	3.69
CarbonDioxide	1.90e-05	9.29e-07	5.76e-08	4.04e-09	4.06	9.23e-04	5.69e-05	4.27e-06	3.08e-07	3.85
SulfurDioxide	3.85e-05	2.26e-06	1.22e-07	7.47e-09	4.11	5.14e-03	2.99e-04	1.64e-05	1.00e-06	4.11
Propane	7.52e-05	3.16e-06	2.20e-07	2.42e-08	3.86	3.54e-03	1.67e-04	1.49e-05	1.74e-06	3.66
Ethane	1.13e-04	6.76e-06	5.94e-07	2.00e-08	4.16	4.88e-03	2.75e-04	2.63e-05	1.02e-06	4.08
n-Butane	1.21e-04	7.91e-06	5.06e-07	2.60e-08	4.06	5.76e-03	3.27e-04	2.77e-05	1.79e-06	3.88

Table 3 and Figure 1 quantify the validation convergence across four successive dyadic grid refinements ($M_{\text{side}} \in \{32, 64, 128, 256\}$, where each 4× increase in N halves the mesh spacing Δx). Across all 15 fluids, halving Δx reduces validation MAPE and MAE by a factor of 12.2× to 17.9×, yielding an effective empirical convergence order $p_{\text{eff}} \in [3.59, 4.16]$. Consequently, refining from 32×32 (1,024 points) to 256×256 (65,536 points) reduces surrogate errors by more than three orders of magnitude ($\sim$ 2,000× to 5,600×) without encountering conditioning breakdown.

To verify physical self-consistency of the underlying dataset, `tests/test_properties.py` executes 47 automated tests across all 15 fluids at $T_{\text{mid}} = (T_{\min} + T_{\max})/2$ and $P = 2.0$ MPa using central finite differences ($\Delta T = 0.01$ K, $\Delta P = 100.0$ Pa), confirming five fundamental thermodynamic identities to within relative tolerances of 10^{-10} to 2×10^{-5}:

$$g_{\text{dep}} = h_{\text{dep}} - Ts_{\text{dep}} = RT\ln\phi,\ \left(\frac{\partial h}{\partial P}\right)_T = v - T\left(\frac{\partial v}{\partial T}\right)_P,\ \left(\frac{\partial s}{\partial P}\right)_T = -\left(\frac{\partial v}{\partial T}\right)_P,\ \left(\frac{\partial \ln f}{\partial P}\right)_T = \frac{v}{RT},\ \left(\frac{\partial h}{\partial T}\right)_P = T\left(\frac{\partial s}{\partial T}\right)_P \tag{10}$$

5 Conclusions, Limitations, and Engineering Implications

The `realgas-ml` v0.3.0 architecture demonstrates that pairing logarithmic pressure normalization and logarithmic target transforms ($\ln Z, \ln\phi$) with local quintic RBF interpolation ($k = 48, m = 2$) reproduces analytical Peng–Robinson and NASA7 thermodynamics across 15 pure fluids to 8–10 significant digits (Max APE $\leq 1.04 \times 10^{-5}$%). By eliminating iterative cubic root finding during forward evaluations and bounding inverse $P(T, v)$ queries within Brent's method, the package provides fast, dependency-free property lookups suitable for process simulation teaching laboratories and offline engineering workflows. Two fundamental engineering caveats documented in `README.md` and `model.py` warrant emphasis. First, because the four targets $[\ln Z, h, s, \ln\phi]$ are interpolated as independent scalar fields, the surrogate's analytical partial derivatives satisfy Equation (10) only up to the $\sim 10^{-8}$ interpolation residual rather than identically; differential sensitivity solvers requiring exact Jacobian symmetry should invoke `peng_robinson.properties` directly or fit a single scalar fundamental potential (such as molar Helmholtz energy $a(T, v)$) and differentiate it analytically. Second, the reported errors quantify numerical fidelity to the PR76 equation of state, not physical accuracy against experimental measurements; PR76 exhibits known structural deviations for strongly polar hydrogen-bonding fluids (H_2O, NH_3) and quantum fluids (He, H_2), where multi-parameter Helmholtz equations of state remain necessary for custody-transfer precision.

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